

Neural Networks & Backpropagation Dynamics

- **Perceptron**: The fundamental biological-inspired building block of artificial intelligence models.
- **Activation Function**: Non-linear transformations such as **ReLU**, **Sigmoid**, and **Softmax** allow models to learn intricate boundaries.
- **Gradient Descent**: An optimization algorithm iteratively tuning network **weights** to minimize empirical loss.
- **Backpropagation**: Efficiently calculates error derivatives from output layer backwards using the chain rule.
- **Overfitting**: Occurs when high model capacity memorizes training noise instead of general patterns.

■ KEY STUDY TERMS FOR DOODLING:

Perceptron | Activation | Gradient | Backpropagation | Overfitting | Weights | Sigmoid | Softmax

Decision Trees, Random Forests & Boosting

- **Decision Tree**: Hierarchical recursive partitioning using **entropy** and information gain splits.
- **Bagging**: Bootstrap aggregation combining multiple independent parallel estimators like **Random Forest**.
- **Boosting**: Sequential ensemble learning where weak learners focus on previous prediction errors.
- **AdaBoost**: Dynamically reweights misclassified sample points across subsequent training rounds.
- **Gradient Boosting**: Fits subsequent base trees directly against the pseudo-residuals of the loss function.

■ KEY STUDY TERMS FOR DOODLING:

Decision | Tree | Ensemble | Boosting | Forest | Entropy | Residuals | Variance

Convolutional Neural Networks (CNNs)

- **Convolution:** Sliding parameterized filter kernels across 2D spatial pixel tensors to detect visual cues.
- **Kernel:** Small matrix of learnable weights tuned to extract edges, textures, and corner gradients.
- **Feature Map:** The resulting activation matrix highlighting specific visual cues at spatial coordinates.
- **Pooling Layer:** Downsamples spatial dimensions using max-pooling or average-pooling to achieve translation invariance.
- **Receptive Field:** The localized sensory patch of original input image influencing a deeper neuron.

■ KEY STUDY TERMS FOR DOODLING:

Convolution | Kernel | Pooling | Filter | Invariance | Pixels | Tensors | Edges

Transformers & Self-Attention Mechanisms

- **Transformer:** Sequence-to-sequence architecture relying entirely on attention without recurrent loops.
- **Self-Attention:** Computes pairwise relevance between token embeddings using Query, Key, and Value vectors.
- **Multi-Head Attention:** Enables representations to jointly attend to information across different subspace perspectives.
- **Tokenizer:** Segments continuous natural language strings into subword tokens and integer vocab IDs.
- **Positional Encoding:** Injects sequential order and positional coordinate information into invariant embeddings.

■ KEY STUDY TERMS FOR DOODLING:

Transformer | Attention | Tokenizer | Embedding | Tokens | Vectors | Encoder | Decoder

Autonomous Agents, Policy & Q-Learning

- **Agent & Environment:** The dynamic interactive feedback loop where an agent observes state and takes action.
- **Reward Function:** Scalar numerical signal guiding an agent toward optimal behavior and goal completion.
- **Policy:** The probabilistic mapping defining an agent's strategy from perceived state to action choice.
- **Q-Learning:** Model-free off-policy algorithm estimating maximum expected future discounted returns.
- **Exploration vs Exploitation:** Balancing trying novel speculative actions versus executing known high-reward actions.

■ KEY STUDY TERMS FOR DOODLING:

Agent | Reward | Policy | Exploration | Environment | State | Action | Discount